

EDA Report: RNA-seq Data Exploration

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For this part of the project (EDA), I focused on exploring the RNA-seq datasets after cleaning to better understand the structure of the data, identify patterns, and check for any issues before moving into deeper analysis.

I started by working with the cleaned gene expression data and metadata from both datasets. After making sure everything was properly aligned by sample ID, I began by looking at the overall structure of the data, including the number of genes, number of samples, and how the variables were distributed. Since RNA-seq data can be highly skewed, I had already applied a log transformation during the cleaning stage, which made the data easier to visualize and interpret.

I then used DESeq2 to begin exploring the data in a more structured way. I created a DESeq2 dataset object by combining the count matrix with the metadata and specifying the variable of interest (such as disease status or severity). DESeq2 is useful because it normalizes the data and accounts for differences in sequencing depth between samples, which makes comparisons more accurate. I used it to generate transformed data (such as variance-stabilized or regularized log values), which are better suited for visualization and exploratory analysis.

One of the main things I explored was the distribution of gene expression values across samples. I used plots to check how the values were spread out and to confirm that the log transformation helped reduce extreme skewness. This step was important to make sure the data is suitable for downstream statistical methods.

I also used PCA (principal component analysis) through DESeq2 to compare samples and see how they cluster. This allowed me to visualize whether samples group together based on biological variables like disease status or severity. From this, I could see whether there are clear differences between groups, which is important for later differential expression analysis.

In addition, I explored the metadata variables themselves. I checked things like the distribution of gender, disease status, and other clinical variables to understand how balanced the dataset is. This is important because uneven group sizes can affect analysis results later on.

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I also examined relationships between variables by creating visualizations that compare gene expression patterns across different groups. For example, I looked at how expression levels differ between disease and control samples, and whether severity levels show any trends. These comparisons helped highlight potential patterns worth investigating further.

Throughout the EDA process, I also checked for any inconsistencies or missing values in both the metadata and expression data. Making sure the data is clean and consistent so there are not any errors in later steps.

Overall, this exploratory analysis helped me better understand the structure and patterns in the RNA-seq data. Using DESeq2 allowed me to properly normalize and visualize the data, and it confirmed that the dataset is ready for more advanced analysis, such as identifying differentially expressed genes.